

LAL 3.2.

$$a) \langle f, g \rangle = \int_1^5 fg \, dx \quad B = \{1, x, x^2\}$$

normalização vetor $\{1\}$:

$$\|1\| = \sqrt{\int_1^5 1 \cdot 1 \, dx} = \sqrt{\frac{16}{3}} = \frac{4}{\sqrt{3}}$$

$$\text{tela } u_1 = \frac{1}{\|1\|} = \frac{1}{2}$$

w_2 ort. pr. x na $\text{span}(u_1)$

$$w_2 = \langle \frac{1}{2}, x \rangle \left(\frac{1}{2} \right) = \frac{1}{2} \int_1^5 \frac{1}{2} x \, dx = x - 3$$

tela $(x - w_2) \perp \text{span}(u_1)$

$x - w_2 = x - 3 \rightarrow$ normalização

$$\frac{x-3}{\|x-3\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 \, dx}} = \frac{x-3}{\sqrt{\frac{16}{3}}} = \frac{\sqrt{3}(x-3)}{4}$$

w_3 ort. pr. x^2 na $\text{span}\left\{\frac{1}{2}, \frac{\sqrt{3}(x-3)}{4}\right\}$ u_2

$$w_3 = \left\langle \frac{1}{2}, x^2 \right\rangle \frac{1}{2} + \left\langle \frac{\sqrt{3}(x-3)}{4}, x^2 \right\rangle \frac{\sqrt{3}(x-3)}{4}$$
$$= \frac{1}{2} \int_1^5 x^2 \, dx + \frac{\sqrt{3}(x-3)}{4} \int_1^5 \frac{\sqrt{3}(x-3)}{4} x^2 \, dx$$

$$= \frac{62}{6} + 6(x-3) \quad \text{potem } (x^2 - w_3) \perp \text{span}\left\{\frac{1}{2}, \frac{\sqrt{3}(x-3)}{4}\right\}$$

normalização $u_3 = \frac{x^2 - \frac{62}{6} + 6(x-3)}{\|x^2 - \frac{62}{6} + 6(x-3)\|} = \frac{x^2 - \frac{62}{6} + 6(x-3)}{\sqrt{\int_1^5 \left(x^2 - \frac{62}{6} + 6(x-3)\right)^2 \, dx}}$

$$= \frac{x^2 - \frac{62}{6} - 6(x-3)}{\sqrt{\frac{256}{45}}} = \frac{\sqrt{45}(x^2 - \frac{62}{6} - 6(x-3))}{16}$$

task OT basis span $\{1, x, x^2\}$ orth to $\langle \cdot, \cdot \rangle$ on
 $\left(\frac{1}{2}, \frac{\sqrt{3}(x-3)}{4}, \frac{\sqrt{45}(x^2 - \frac{62}{6} - 6(x-3))}{16} \right)$

b) orth. pr. $\sin x$ on span $\{1, x, x^2\}$.

$$\begin{aligned}
 &= \left\langle \frac{1}{2}, \sin x \right\rangle \frac{1}{2} + \left\langle \frac{\sqrt{3}(x-3)}{4}, \sin x \right\rangle \frac{\sqrt{3}(x-3)}{4} \\
 &+ \left\langle \frac{\sqrt{45}(x^2 - \frac{62}{6} - 6(x-3))}{16}, \sin x \right\rangle \frac{\sqrt{45}(x^2 - \frac{62}{6} - 6(x-3))}{16} \\
 &= \frac{1}{2} \int_1^5 \frac{\sin x}{2} dx + \frac{\sqrt{3}(x-3)}{4} \int_1^5 \frac{\sqrt{3}(x-3)}{4} \sin x dx \\
 &+ \frac{\sqrt{45}(x^2 - \frac{62}{6} - 6(x-3))}{16} \int_1^5 \frac{\sqrt{45}(x^2 - \frac{62}{6} - 6(x-3))}{16} \sin x dx \\
 &= \cancel{-0.122x^2 + 1.379x + 10.10} + 0.122x^2 + 0.729x + 0.203
 \end{aligned}$$

c) $\| \sin x + 0.122x^2 - 0.729x - 0.203 \|$

$$= \int_1^5 \left(\sin x + 0.122x^2 - 0.729x - 0.203 \right)^2 dx = 2.622$$